Figure 2: Riak TS database—Web interface

Features

- Supports addition of new nodes to the existing cluster architecture without sharding; data is automatically and uniformly distributed across the database cluster.
- Supports DDL or Data Definition Language for table and field definitions, and supports storage of both structured and semi-structured data.
- Supports multi-cluster replication, which facilitates systems administrators to replicate the data across the in-house data centre and any geo-location data centre anywhere in the world.
- Supports SQL-like data queries by users for easy and flexible access to global databases.
- Supports application integration with APIs and client libraries in various languages like Java, Ruby, Python, Erlang, Go, Node.js and .NET.
- Riak Meso framework provides efficient cluster resource management and 'push button' scale-up/down for RIAK nodes.
- Supports full integration with Apache Spark for operational analysis of time series data.

  Official website: *http://basho.com/products/riak-ts/*
  Latest version: 1.3

MongoDB

MongoDB is a highly powerful, flexible, free and open source, document-oriented, scalable and general-purpose database. It has the ability to scale out features such as secondary indexes, range queries, sorting, aggregations and geospatial indexes. It is classified as a NoSQL database as it uses JSON-like documents with schemas.

MongoDB adds dynamic padding to documents and pre-allocates data files to trade extra space usage for consistent performance. It makes efficient use of RAM for caching and correcting queries for indexes. MongoDB supports a rich query language to support read and write operations (CRUD) as well as data aggregation, text search and geospatial queries.

Features

- Supports generic secondary indexes for a variety of fast queries, and provides unique, compound, geospatial and full-text indexing features to users.
- Supports 'aggregation pipelines' to build complex aggregations from simple pieces for optimisation of the database.
- Supports TTL (Time-To-Live) collections for data that should expire after a certain period of time.
- Supports easy-to-use protocol for storing large files and metadata files.
- Supports JSON to store and transmit information. JSON, being standard protocol, is a great advantage for both the Web and the database.
- Supports Map-Reduce on the server side for information processing using JavaScript functions.
- Supports MongoDB Management Service (MMS) tool for allowing users to track databases and backing up the data.
- Supports automatic load balancing configuration because of data placed in shards.

  Official website: *https://www.mongodb.com/*
  Latest version: 3.4

RethinkDB

RethinkDB is an open source, distributed database primarily used to store JSON documents; it has the capacity of scaling up to multiple machines. RethinkDB is regarded as the first and foremost choice for developers, especially IoT based developers, for feeding real-time data. It has completely revolutionised the traditional database architecture by invoking a new access model to update query results to applications in real-time. RethinkDB offers a flexible query language for monitoring APIs, and is highly easy to set up and learn.

RethinkDB offers a number of advantages over MongoDB. These are:

- An advanced query language that supports table joins, sub-queries, and massively parallelised distributed computation.
- An elegant and powerful operations and monitoring API that integrates with the query language, and makes scaling RethinkDB dramatically easier.
- A simple and beautiful administration UI that lets you shard and replicate in a few clicks, and offers online documentation and query language suggestions.

Features

- *Fault tolerance:* It supports the automatic shift to a new server if the primary server fails.
- *Easy addition of nodes:* Plug-and-play of nodes in real-time, without any downtime for even a single second.
- *Asynchronous application programming interfaces:* Supports asynchronous queries via Eventmachine in Ruby and Tornado.